

Commercial building indoor environmental quality models: A critical review



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ABSTRACT

Indoor environmental quality (IEQ) models merge various influencing IEQ factors into a single overall indicator. The outcome of the IEQ model is usually a numerical rating or score that provides an evaluative summary of the IEQ performance. Many publications have developed IEQ models from their field measurement and/or satisfaction survey results, however with distinct measurement protocols, IEQ weighting schemes and analytical methods. This paper presents a comprehensive review of 25 studies focusing on developing IEQ evaluation models in commercial buildings. Two IEQ measurement approaches (objective and subjective measurement), and three types of IEQ models—subjective-objective models, objective models and subjective models—are first differentiated. Five commonly used analytical methods (linear regression, non-linear regression, analytical hierarchy process, ensemble averaging method, and updated IEQ acceptance model) are discussed. The interaction effects between IEQ factors and relative contribution of them to the overall IEQ satisfaction are also assessed. Twenty-five studies are then mapped by data collection method, analytical method and number of survey responses. Five key criteria for developing quality IEQ models—number of IEQ factors included, spatial and temporal variability in physical measurement, analytical methods for model development, and the number of responses if subjective surveys are carried out—are proposed and discussed. A simple ranking and scoring system is propounded for evaluating the quality of the IEQ models developed in the reviewed studies. Results show that two studies achieved a total score equal to or higher than 80% of the total score, and 16 studies achieved a minimum of 60% of the total score. This tool can help researchers quickly assess the accuracy and reliability of IEQ models developed from diverse methods and make an informed decision.

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1. Introduction

Approximately 90% of human life is spent indoors [1]. Indoor environmental quality (IEQ) relates to building performance from the perspective of its occupants and includes their perception of thermal comfort, acoustics, lighting, and indoor air quality (IAQ), among other things [2]. Previous studies demonstrate that for any enterprise, the personnel costs are orders of magnitude higher than the buildings' operational costs and improving IEQ can reduce the number of absenteeism and complaints, and elevate workplace productivity [3,4]. Although the relationship between IEQ and productivity remain contentious [5], it is agreed that more satisfied

workforce in a good indoor environment generally brings better outcomes for the organizations [6,7].

Given that IEQ has different components (e.g., thermal comfort, IAQ, lighting and acoustic comfort), a summative indicator is needed to represent the buildings' overall indoor environment condition merging all influencing IEQ factors. This is usually a numerical rating or score that provides an gauging numerical summary of the individual IEQ factor performance data [8]. In this paper, the summative indicator is named as "IEQ models", although different terms have been used by the literature to fulfil the same/similar function, e.g., IEQ rating system, IEQ index, and IEQ score.

The IEQ performance data used to generate the IEQ models can come from two major sources [9]: objective measurements (e.g., room temperature, relative humidity, CO₂, illuminance, noise level, etc.) and subjective surveys (satisfaction questions regarding noise, lighting, thermal comfort, etc.). IEQ models combined the collected

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Nomenclature

Acronyms

AC	Acoustic comfort
AHP	Analytical hierarchy process
BREEAM	Building research establishment environmental assessment method
DGNB	Deutsche Gesellschaft für Nachhaltiges Bauen
HVAC	Heating, ventilation, and air conditioning
IAQ	Indoor air quality
IEQ	Indoor environmental quality
ITACA	Istituto per l'innovazione e trasparenza degli appalti e l'compatibilità ambientale
LEED	Leadership in energy and environment design
LiderA	Lead for the environment in search of sustainability construction
MNS	Maximum number of surveys
NABERS	National Australian built environment rating system
POE	Post occupancy evaluation
PRISMA	Preferred reporting items for systematic reviews and meta-analyses
TC	Thermal comfort
VC	Visual comfort

Symbols

f	Time Fraction Weighted Mean Vector
F	Time fraction matrix
FS	IEQ factors score
E	Illumination intensity
L_A	Sound pressure level
P	Collected points
S	Satisfaction rate
t_o	Operative temperature
w	Relative weight
W	Weight factor

Subscripts

AM	Analytical method
IIF	Included IEQ factors
SD	Spatial diversity
SR	Survey responses
TD	Temporal diversity

data of objective measurements and/or subjective surveys and summarized them into a rating, which can then be used for comparison and benchmarking purposes, such as in green building rating schemes and international building standards. The accuracy and validity of such IEQ models, therefore, depends on the measurement devices, measurement protocols, methods used to weight IEQ factors and develop the quantitative relationship.

In a comprehensive review, Heinzerling et al. [8] thoroughly discussed the methods and tools used for subjective and objective measurement, highlighted the lack of common IEQ measurement protocols and consideration of interaction effects between IEQ factors in the IEQ models, and also proposed new assessment class and IEQ weighting schemes. Almost ten years passed since their review. There are recent studies that have explored new research topics, such as the inclusion of interaction effects in IEQ models [10], updated IEQ acceptance model [11], and justification of IEQ models [12]. In addition, there is a lack of a simple and quick tool for future researchers to check the quality of the IEQ models developed based on diverse methods, even though Heinzerling et al. [8] have mentioned some important features that quality IEQ models should possess. Therefore, the current systematic review study aims to fill the above gaps by 1) identify, synthesize, and summarize new and important research topics in IEQ modelling, 2) identify key criteria for developing quality IEQ models, and 3) provide a simple ranking and scoring tool to assess the quality of the IEQ models.

This research is structured in five sections. Section 2 presents the methodology for the current systematic literature review. Section 3 summarizes various research topics and methods for IEQ modelling. Section 4 then discusses the IEQ model evaluation criteria and propounded a simple tool to assess IEQ model quality. Finally, conclusions are presented in Section 5.

2. Methodology

The systematic quantitative literature review method allows a reliable assessment of a specific research topic and to identify the research gap for future research. In this study, the preferred reporting items for systematic reviews and meta-analyses

(PRISMA) technique was applied to organise the literature for the review process [13]. PRISMA is a four-step systematic data collection process (identification, screening, eligibility, and records inclusion). Similar studies have adopted this approach for their review process and content analysis [14,15]. Fig. 1 explains the four-step process via the PRISMA approach.

2.1. Identification

The related records in this phase were selected by including only peer-reviewed journals and conference publications. An unrestricted search was initially performed on Scopus and Web of Science, which are well-known as the most reliable database engines for academic research to find relevant publications in the field of building IEQ modelling. The relevant keywords were obtained based on an investigation of publications and the authors' knowledge.

The data structure in these search engines includes publication types, authorships, journal or conference titles, keywords, abstracts, institutions, and references. Keywords were input in Scopus search engine using the TITLE-ABS-KEY as follows: ("indoor environmental quality" OR IEQ) AND (model* OR index* OR weighting). By adding "*" at the end of each word, related words (like modelling or indexing) would be included in the search results. The initial search resulted in 233 publications in Scopus and 822 publications in Web of science databases. Therefore, 1,055 papers were retrieved from an unrestricted search.

2.2. Screening

The publications collected in the previous stage were filtered using the following inclusion/exclusion criteria.

1. Only publications in the past two decades (2001–2021) were included to examine the recent development of IEQ models.
2. Subject area was limited to engineering, computer science, social sciences, environmental science, energy, and decision making. Fields like medicine and business were not relevant to the study and were excluded.

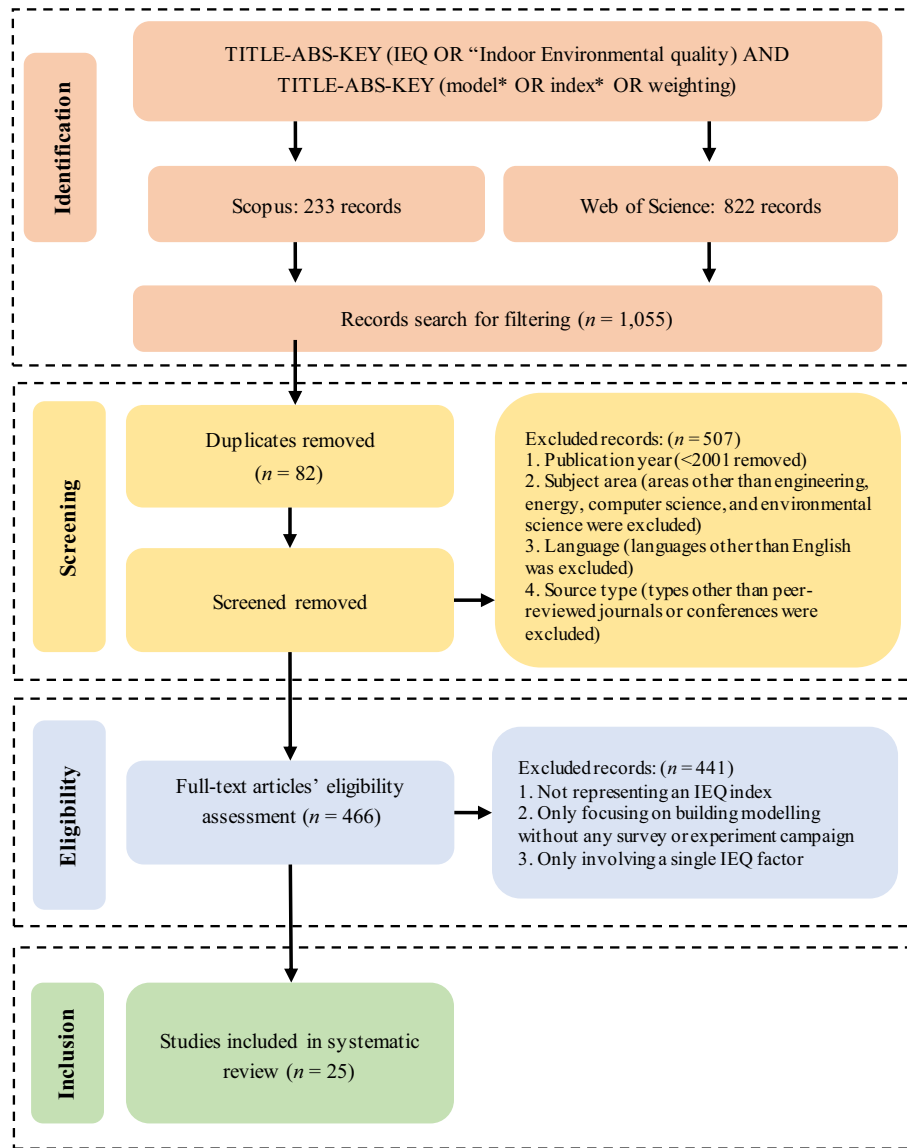


Fig. 1. Systematic literature review process.

3. Papers not in English were removed.
4. Sources of publication was restricted to conference and journal articles.

Also, 82 duplicate papers were removed. Therefore, 466 articles remained for the eligibility phase.

2.3. Eligibility

Among 466 articles, the ones that simultaneously meet the following three criteria were retained in this study.

1. Representation of IEQ model, index or weighting systems
2. Inclusion of occupant surveys and/or physical measurement
3. Evaluation and comparison of impact of various IEQ factors and not only focusing on a single factor.

Finally, after passing the eligibility test, 25 articles were selected for inclusion in the study to be further studied.

2.4. Thematic analysis

Twenty-five selected publications were further investigated to gain knowledge about IEQ modelling. These studies have adopted different research methods, including controlled laboratory experiments, field monitoring in spaces and buildings, and occupant surveys, among other things. Thematic analysis was adopted to identify, analyse, and construct themes and topics discussed in these papers. The following section will present the following main topics: IEQ measurement approaches, type of IEQ models, analytical methods, IEQ model justification, and the interaction between and relative importance of IEQ models.

3. Results

The present study mainly focused on the studies published after Heinzerling et al. [8], however a few older papers were also included to investigate the research trends. The main information covered in these 25 studies are presented in Table 1. Nine IEQ factors were evaluated in the extracted studies. Thermal comfort, visual comfort, acoustic comfort, and IAQ were four main factors

Table 1

Published research related to IEQ models in commercial buildings.

#	Ref.	Year	Sample size	Duration	TC	IAQ	VC	AC	Privacy	Furnishing	Personal control	Cleanliness and maintenance	Available space	First-hand data	Secondary data	Measurement	Surveys
1	[24]	2002	12	–	✓	✓	✓	✓	–	–	–	–	–	✓	–	✓	–
2	[12]	2005	4655	–	✓	✓	✓	✓	–	–	–	–	–	–	–	–	✓
3	[25]	2005	422	Winter and summer	✓	✓	✓	✓	–	–	–	–	–	✓	–	–	✓
4	[26]	2008	293	One-off survey	✓	✓	✓	✓	–	–	–	–	–	✓	–	–	✓
5	[35]	2009	422	–	✓	✓	✓	✓	–	–	–	–	–	✓	–	–	✓
6	[30]	2009	–	One-off survey	✓	✓	✓	✓	–	–	–	–	–	–	–	–	✓
7	[27]	2012	500	Different seasons	✓	✓	✓	✓	–	–	–	–	–	✓	–	–	✓
8	[31]	2012	43,021	One-off survey	✓	✓	✓	✓	✓	✓	✓	✓	✓	–	✓	–	✓
9	[34]	2012	–	1 year	✓	✓	✓	✓	–	–	–	–	–	✓	–	✓	–
10	[9]	2012	70	2 days	✓	✓	✓	✓	–	–	–	–	–	✓	–	✓	✓
11	[8]	2013	52,980	–	✓	✓	✓	✓	–	–	–	–	–	–	–	–	✓
12	[28]	2015	1000	8 months	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	–	✓	✓
13	[33]	2017	–	–	✓	✓	✓	✓	–	–	–	–	–	–	–	–	✓
14	[36]	2018	1050	Different seasons	✓	✓	✓	✓	✓	–	✓	✓	✓	✓	✓	✓	✓
15	[11]	2018	730	One-off survey	✓	✓	✓	✓	–	–	–	–	–	–	✓	–	✓
16	[16]	2018	496	2 months	✓	–	✓	–	–	–	–	–	✓	–	–	✓	✓
17	[29]	2020	367	2 weeks	✓	✓	✓	✓	✓	–	✓	–	✓	✓	–	✓	✓
18	[10]	2020	248	2 months	✓	✓	✓	✓	–	–	–	–	–	✓	–	✓	✓
19	[19]	2020	572	1 year	✓	✓	✓	✓	–	–	–	–	–	✓	–	✓	✓
20	[20]	2020	300	One-off survey	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	–	–	✓
21	[21]	2020	43	1 month	✓	✓	✓	✓	–	–	–	–	✓	✓	–	✓	✓
22	[32]	2020	–	–	✓	✓	✓	✓	–	–	–	–	–	–	✓	–	✓
23	[22]	2021	666	One-off survey	✓	✓	✓	✓	✓	✓	✓	✓	✓	–	–	–	✓
24	[23]	2021	126	1 day	✓	✓	✓	✓	–	–	–	–	–	✓	–	✓	✓
25	[17]	2021	234	1 day	✓	✓	–	–	–	–	–	–	–	✓	–	✓	✓
Total					25	24	24	23	6	4	7	5	8	18	10	15	23

TC: thermal comfort; IAQ: indoor air quality; VC: visual comfort; AC: acoustic comfort

included in most studies. Also, there were five perceptual factors (privacy, furnishing, personal control, cleanliness and available space) which were considered in only a few studies. All publications considered thermal comfort, while one paper did not include IAQ [16]; another research excluded visual comfort [17]. Surprisingly, two studies did not include acoustic comfort when generating their IEQ models [16,17], albeit previous studies highlighted the crucial role of acoustic comfort in achieving IEQ satisfaction [18]. Privacy, furnishing, personal control, cleanliness and maintenance, and available space were factors commonly considered in IEQ models built on satisfaction surveys. Nine publications considered these perceptual factors in IEQ models. While most papers collected first-hand data in their studies [9,10,16,17,19–29], some publications developed IEQ models based on data extracted from previous studies [8,11,12,30–33]. Also, there are a few papers which combined the first-hand and secondary data to develop an IEQ model [34–36]. Section 3.1–3.5 evaluate and compare 25 extracted publications from different aspects.

3.1. IEQ measurement approaches

3.1.1. Objective measurement

Using calibrated and accurate measuring devices is essential for developing IEQ models. In recent decades, many instruments and tools have been used for IEQ measurement. Adoption of specific tools and methods is heavily dependent on the research aims of the study. Carrying out objective IEQ measurement is usually labour and cost intensive. For instance, based on the research goals, a representative time and location(s) must be selected for IEQ monitoring; test instruments must be calibrated; and experts should spend time installing the measuring devices, collecting data, and removing instruments afterwards.

Although all the reviewed studies presented their research method as an 'IEQ experiment', there is no standard data collection

procedure to gather an illustrative sample. A wide range of spatial and temporal resolutions have been used in IEQ monitoring studies (see Table 2). Apart from that, there are other things to consider when determining the sensor locations, such as the occupancy conditions, exposure of sensors to direct sunlight/ air path/ personal CO₂ bubbles, and potential safety hazards when cables are in use. Currently, there is no standard objective IEQ parameters measurement protocol that can be followed by researchers.

3.1.2. Subjective measurement

Occupant satisfaction, well-being, and productivity surveys constitute the easiest and most affordable method of IEQ evaluation in buildings [37]. Although many standards dictate what an acceptable indoor environment is and how to measure it, occupant satisfaction is of utmost importance among building managers. Most of IEQ surveys in extracted papers were in written form; however, online surveys are becoming increasingly popular in recent years. However, Galatioto et al. [38] stated that surveys administered in the written form generally have a higher response rate (80%) in comparison with online questionnaires (30%).

The difficulty of finding a representative period, interpreting the results, and selecting the right questions to ask are other issues concerning surveys [8]. The answers that respondents offer in a survey are greatly affected by the phrasing of the questions. The inappropriately composed survey questions may cause biased or mistaken results. Some additional factors (e.g., ethnicity, physiological states, and socio-economic states) are not usually considered in surveys.

3.2. Type of IEQ models

The main goal of IEQ models is to evaluate indoor environmental quality conditions based on standard benchmarks or occupant satisfaction. Many weighting models have combined satisfaction

Table 2
Summary of c in previous studies.

	Temporal/ Spatial resolutions	Thermal Comfort	IAQ	Visual Comfort	Acoustic Comfort
[16]	Temporal	Continuous measurement in summer and winter Automatic recording of Thermal and visual parameters at one-minute intervals Manual measurement of a weighted sound level and CO ₂ concentration every five minutes			
[17]	Spatial	Installing fixed sensors in the vicinity of the 32 occupants			
	Temporal	One-off measurement for two minutes	–		–
[21]	Spatial	Installing mobile sensors within a distance of 1.5 m to surveyed participants' workstations		–	–
	Temporal	Continuous measurement for more than a month in summer and winter Automatic recording at 10-minute intervals			
[10]	Spatial	Placement of IEQ monitors and sensors in representative open office zones at 0.75 m above the floor			
	Temporal	Continuous measurement for more than a month			
[34]	Spatial	Three fixed locations	Three fixed locations	Eight fixed locations	Two fixed locations
	Temporal	Daily measurements with constant intervals of 15 min for a whole year based on work hours			
	Spatial	Selecting measurement points in the middle of two offices with the area of 11 m ² and 40 m ²			

scores for IEQ sub-factors to arrive at an inclusive IEQ satisfaction score [8]. Most of the studies developed the IEQ model by linear or non-linear regression analysis. IAQ, thermal comfort, visual comfort and acoustic comfort are the most commonly included sub-factors in IEQ modelling studies. By multiplying each IEQ sub-factor score (FS_i) with its weight (W_i), the IEQ_{model} can be calculated as (Eq. (1)).

$$IEQ_{model} = \sum_{i=1}^n W_i FS_i \quad (1)$$

The IEQ models fall into three categories: subjective-objective models, objective models, and subjective models, as shown in Table 3. Two former categories were introduced by Heinzerling et al. [8], while some recent research adopted the third category. Three IEQ model categories are briefly compared and discussed below.

3.2.1. Subjective-objective models

These models are utilised in studies attempting to find correlation between subjective and objective IEQ measures and provide equations that predict occupant satisfaction for each IEQ category, e.g., thermal environment, IAQ. The IEQ models that consider both objective measurement and subjective survey were initiated by Mui et al. [25]. They developed the first IEQ logger to expedite the continuous monitoring and adopted a linear regression model for determining the dissatisfaction rate with IEQ based on the rate of dissatisfaction with thermal comfort, IAQ, aural comfort, and visual comfort. Based on the initial regression, the impact of visual comfort was discovered to be irrelevant to total satisfaction, while thermal comfort had the greatest impact.

The overall IEQ satisfaction can also be stated as a function of various IEQ parameters, such as indoor temperature, air speed, relative humidity, CO₂ concentration, PM_{2.5} concentration [9,27]. Such overall IEQ model result is then compared to a fixed range of standard IEQ parameter values. Cao et al. [27] used a two step-method for their IEQ model development (Eq. (2)).

$$IEQ = c_0 + c_1 \times S_{TC} + c_2 \times S_{IAQ} + c_3 \times S_{VC} + c_4 \times S_{AC} \quad (2)$$

Using 500 satisfaction survey responses, Cao et al. determined the coefficients (C_0 – C_4) in multivariate linear equation. In the second step, they calculated the relationship between each IEQ factor satisfaction and its corresponding physical parameter (operative temperature t_o , CO₂ concentration C_{CO_2} , sound pressure level L_A and illumination intensity E) to establish the IEQ model (Eq. (3)).

$$\begin{aligned} IEQ = & 0.075 + 0.316 \times (-0.0063t_o^2 - 0.287t_o - 2.934) \\ & + 0.118 \times (-0.0002C_{CO_2} + 0.244) + 0.171 \\ & \times (-5 \times 10^{-7}E^2 + 0.0011E - 0.106) + 0.224 \\ & \times (-0.023L_A - 1.382) \end{aligned} \quad (3)$$

Table 3
Comparison of three types of IEQ models in the literature [8].

Phase	Subjective-Objective models	Objective models	Subjective models
Data Type	Objective measurement & Satisfaction surveys	Objective measurement	Satisfaction surveys
IEQ model	development	Regression between objective data and subjective surveys to predict occupant satisfaction for a specific IEQ factor	Compare measurement to fixed set of standard values, and determine classification
Regression between overall IEQ	satisfaction and satisfaction towards each IEQ factor Combine sub-indexes into an overall IEQ score	Determine time spent in each assessment classification to calculate an index for an IEQ factor Combine sub-indexes into an overall IEQ score	
Evaluation	Comparison of IEQ index with fixed standard ranges Comparison of IEQ index in different building types	Compare IEQ index with fixed standard ranges	Comparison of IEQ index in different building types

3.2.2. Objective models

Some studies built their IEQ models based on objective measurements and compared collected data with a fixed set of standard values that reveals what assessment classification the measurement fell within. The length of time spent in each assessment classification can then be determined and used to calculate a single index for each IEQ factor [34]. These sub-indexes are then integrated into an overall IEQ score through the weighting process. While such studies may include satisfaction surveys, the collected subjective surveys are not involved in IEQ model development. Marino et al. [34] adopted the criteria specified by EN 15,217 standard and defined four IEQ quality categories (high level of expectation, normal level of expectation, acceptable and not acceptable). They defined a matrix representing time fraction $[F]$ in each IEQ quality category (Eq. (4)). The number of rows and columns signi-

fies the number of monitored IEQ parameters and number of IEQ quality categories, respectively.

$$[F] = \begin{Bmatrix} F_{1,I} & F_{1,II} & F_{1,III} & F_{1,IV} \\ F_{2,I} & F_{2,II} & F_{2,III} & F_{2,IV} \\ \dots & \dots & \dots & \dots \\ F_{n,I} & F_{n,II} & F_{n,III} & F_{n,IV} \end{Bmatrix} \quad (4)$$

Marino et al. [34] put together the relative weights of IEQ factors introduced by Bluyssen et al. [39], and structured Relative Weight Vector $\{w\}$ of monitored IEQ parameters (Eq. (5)).

$$\{w\} = \begin{Bmatrix} 0.173 \\ 0.189 \\ 0.181 \\ 0.150 \\ 0.146 \\ 0.160 \end{Bmatrix} \quad (5)$$

Then, Time Fraction Weighted Mean Vector $\{\bar{f}\}$ was calculated by multiplying the transpose of time fraction matrix by relative weight (Eq. (6)).

$$\{\bar{f}\} = [F]^T \{w\} = \begin{Bmatrix} \bar{f}_I \\ \bar{f}_{II} \\ \bar{f}_{III} \\ \bar{f}_{IV} \end{Bmatrix} \quad (6)$$

Finally, a unique IEQ index was developed based on the Time Fraction Weighted Mean Vector.

$$IEQ = 100\bar{f}_I + 70\bar{f}_{II} + 35\bar{f}_{III} \quad (7)$$

Such methods are only based on objective measurement values, the assessment class informed by related international standards, and time spent in each assessment class.

3.2.3. Subjective models

In studies that utilise subjective evaluation models, participants' opinions towards IEQ would be investigated by online or in-person surveys. Then, using different regression methods, the relationship between each IEQ factor and overall IEQ satisfaction can be predicted. Cheung et al. [22] and Kim and de Dear [31] developed these subjective models in their studies.

Cheung et al. [22] built an IEQ survey with 18 satisfaction questions on a 7-point Likert scale. They collected 666 individual office occupants' responses and calculated dissatisfaction rate and satisfaction rate for each IEQ factor based on Eq. (8) and Eq. (9).

Dissatisfaction rate

$$= \frac{(\text{very dissatisfied} + \text{dissatisfied} + \text{somewhat dissatisfied})}{\text{Total responses}} \quad (8)$$

Satisfaction rate

$$= \frac{(\text{very Satisfied} + \text{Satisfied} + \text{somewhat Satisfied})}{\text{Total responses}} \quad (9)$$

Then, they used linear regression method to find the relationship between total occupant satisfaction/dissatisfaction rate and occupant satisfaction with 18 IEQ factors. Akaike information criterion was employed to exclude less important factors. The final IEQ model (Eq. (10)) contains occupants' satisfaction scores (denoted by S) with 9 IEQ factors.

$$\begin{aligned} IEQ = & 0.02 + 0.24 \times S_{\text{Cleanliness}} + 0.15 \times S_{\text{Noise}} + 0.13 \\ & \times S_{\text{Furnishing}} + 0.11 \times S_{\text{Electriclight}} + 0.10 \times S_{\text{Stiffness}} + 0.10 \\ & \times S_{\text{Humidity}} + 0.09 \times S_{\text{Personalcontrol}} + 0.08 \times S_{\text{Viewsfromwindow}} \\ & + 0.07 \times S_{\text{Dresscode}} \end{aligned} \quad (10)$$

Number of buildings and survey respondents may have influence on the accuracy of such analysis, and it needs a large database. Researchers used different analytical methods to generate IEQ models. In the next section, different methods are discussed.

3.3. Analytical methods

As mentioned, IEQ models can be developed based on objective measurements and/or subjective surveys [40]. A purely objective IEQ model is typified by statistical models, complicated calculation procedure, and an intensive data collection [41]. In comparison, a complete subjective IEQ modelling method depends on criteria weights informed directly by occupants or experts via interviews or questionnaires. Fig. 2 illustrates the distribution of analytical methods among the extracted studies.

3.3.1. Linear regression analysis

The most well-developed weighting approach is linear regression. Most of the studied papers in this research used linear regression analysis to determine IEQ weightings. Mui and Chen were the pioneer team who used linear regression in their IEQ weighting studies [25]. Their IEQ model predicts the percentage of dissatisfaction towards IEQ from operative temperature, CO₂ concentration and sound pressure level.

Humphreys [12] also combined occupants' satisfaction with individual IEQ factors into a single index. Satisfaction data were collected from 4,655 occupants in 26 offices in five European countries. In the combined IEQ index, warmth and air quality were introduced as more influential sub-factors of occupant satisfaction than the impact of lighting and humidity. He concluded that satisfaction with individual IEQ factors does not guarantee overall IEQ

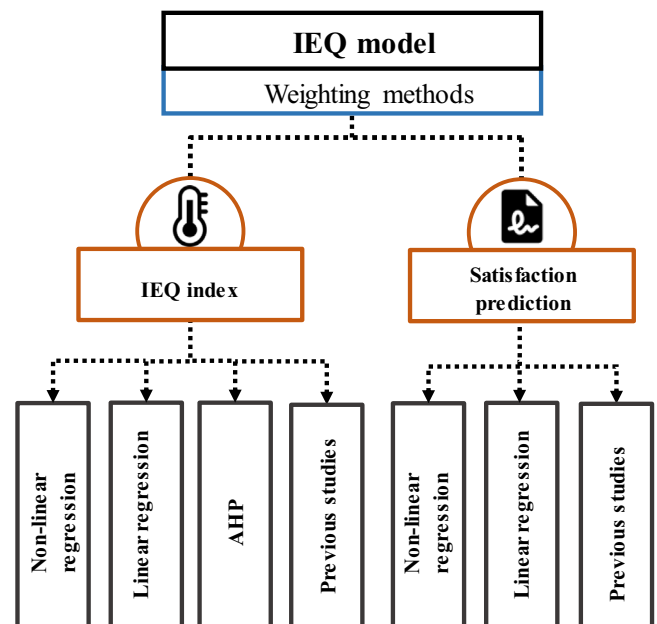


Fig. 2. Categorization of IEQ model weighting methods among reviewed studies.

satisfaction. Likewise, dissatisfaction with individual IEQ factors does not preclude overall IEQ satisfaction.

Cao et al. [27] conducted IEQ monitoring in public buildings in Beijing and Shanghai in 2008 and 2009. Their research included measurements of IEQ factors, e.g., indoor temperature, mean radiant temperature, relative humidity, air speed, CO₂ concentration, illumination, and weighted sound pressure levels. Also, they investigated occupants' satisfaction with each IEQ factor. The relation between each IEQ parameter values and occupant satisfaction with this IEQ parameter was determined by using the least squares method. Then, weighting factors were discovered by using multivariate linear regression of occupants' satisfaction with each IEQ factor and the overall IEQ satisfaction.

Ncube and Riffat [9] introduced a rapid IEQ assessment model for air-conditioned offices in the UK. They included building design, measurements, and surveys for model development. The authors highlighted that represented model properly calculated the behaviour of the dependent variable ($R^2 = 0.94$). Although a comparison with a previously conducted analytical hierarchy process (AHP) analysis by Chiang et al. [42] is presented as model validation, the fundamental differences between the case studies (an office building and a senior citizen care centre) were neglected. Moreover, the measurement method was not consistent during research. For instance, the data collection time step varied between the two studied buildings, and measurements were taken in the heating season in one building, and in the cooling season in the other building.

Heinzerling et al. [8] explored subjective and objective office building IEQ models and discussed the correlations. Also, they correlated the satisfaction data of 52,980 surveys in Frontczak et al. [43] and developed a weighting scheme for office building IEQ benchmarking. Xu et al. [16] introduced space factor into IEQ modelling studies. They evaluated the effects of building space, thermal comfort, and visual comfort on overall IEQ satisfaction. Their results revealed that when temperature, building space, and illumination match the IEQ design standards, an occupant's satisfaction with the building space (followed by illumination and thermal comfort) has the most considerable impact on overall IEQ satisfaction. Table 4 shows the developed linear regression models in extracted papers.

3.3.2. Non-linear regression analysis

Non-linear regression analysis was used in many previous studies. Common methods include logistic regression, arithmetic mean, and geometric mean models (Table 4). Multivariate regression in IEQ models was initiated by Hong Kong Polytechnic University researchers [44]. IEQ satisfaction was regressed against occupants' satisfaction with specific IEQ factors. Lai et al. [44] used a two-step method to correlate IEQ satisfaction with physical measurements. In the first step, the relationship between each objective IEQ parameters and satisfaction with individual IEQ factors was calculated, and in the second step, a non-linear regression was conducted to determine the relationship between individual satisfaction and overall IEQ satisfaction. Thermal and acoustic comfort were found to be the most influential factors, whereas IAQ was deemed as the least significant.

Based on the evaluations of 293 office occupants, Wong et al. [26] introduced an empirical model to estimate the overall IEQ acceptance (Table 4). It was calculated by a multivariate logistic regression model containing four main IEQ factors. Their relative significance, ranked from most to least important, was indoor thermal comfort, IAQ, noise level, and visual comfort. Also, the authors used this model to benchmark air-conditioning offices in Hong Kong based on the predicted overall IEQ acceptance. Mui et al. [35] used the IEQ acceptance procedure in Wong et al. [26] to benchmark offices in a five-star rating scheme. Developed model

used an adaptive thermal comfort approach for various occupants' metabolic rates in a conventional office environment. In other words, occupants' clothing type would be adapted for the perceived thermal condition.

Tang et al. [10] introduced a non-linear IEQ model based on the geometric mean of the individual IEQ factor satisfaction scores (Table 4). The geometric mean will always be lower than the arithmetic mean except when all factors have equal values. Only in that case will the former equate to the latter. Results demonstrated that the geometric mean model had higher prediction accuracy than the linear regression model, evidenced by a 14% reduction of residual standard error.

Other IEQ weighting studies found in this research employed the non-linear method proposed by Lai et al. [44] to evaluate overall IEQ satisfaction. Fassio et al. [45] used the equations proposed in Wong et al. [26] to calculate individual IEQ satisfaction. However, the insufficient participants in the research led to inconsistent results, and it was mentioned that at least 300 occupants are required to obtain optimal statistical reliability [45].

3.3.3. Ensemble averaging method

Wei et al. [32] compared the average weightings of the four main IEQ factors (VC, IAQ, AC, and VC) in seven green building rating certificates (LEED, KLIMA, BREEAM, DGNB, LiderA, ITACA, and NABERS) with those proposed in nine research papers published between 2002 and 2018 that have derived an overall IEQ rating (Table 4). The results indicate that while the importance of thermal and visual comfort in both groups is almost the same, green building standards assigned a relatively higher weighting to IAQ in comparison with published papers.

3.3.4. Analytical hierarchy process (AHP)

Expert panels and analytical calculation constitute another procedure that determines the influence of different IEQ factors. AHP is a technique to create a suitable weighting system by prioritising and allocating relative weightings for the evaluation based on a group of experts' perceptions. This method has been utilised for different purposes, such as determination of building refurbishment priorities [46], building sustainability assessment [47], and green building certificate selection [48].

In Chiang and Lai [24], twelve experts (including five architects, three professional engineers, and four university professors) were consulted to develop a comprehensive IEQ index. It was believed that a combined effect accumulated from each category of physical environmental impact on occupants' health and well-being. Therefore, the IEQ index was designed based on the summation of the evaluated score of the physical environment, multiplied by their weighting.

Online surveys of building experts (engineers, architects, and planners) in the UK and China captured their opinions on green buildings' indoor environment and sustainability [49]. All the expert groups in the UK believed that the most important IEQ factor should be thermal comfort, and the IAQ had also been emphasised by all expert groups [49]. In contrast, in China, engineers considered all the IEQ factors to be equally important, while experts with planning and design backgrounds agreed that IAQ had the highest importance [49].

Miao and Ding [50] invited 60 experts in the construction industry to evaluate the importance of acoustics, luminosity, thermal comfort, and IAQ in public buildings. Then, the AHP method was employed to determine the weights. Based on the experts' opinions, thermal comfort and IAQ accounted for the highest importance in total IEQ satisfaction.

Table 4
Summary of IEQ models in reviewed papers.

Study	Method	R ²	IEQ Prediction model
[23]	Multivariate linear regression	0.92	IEQ = 0.285 × TC + 0.35 × IAQ + 0.170 × VC + 0.195 × AC
[25]	Multivariate linear regression	–	IEQ = 0.42 × TC + 0.09 × IAQ + 0.28 × AC
[27]	Multivariate linear regression	0.46	IEQ = 0.316 × TC + 0.118 × IAQ + 0.171 × VC + 0.224 × AC + 0.075
[8]	Multivariate linear regression	–	IEQ = 0.12 × TC + 0.2 × IAQ + 0.29 × VC + 0.39 × AC
[9]	Multivariate linear regression	0.94	IEQ = 0.3 × TC + 0.36 × IAQ + 0.16 × VC + 0.18 × AC
[10]	Multivariate linear regression	0.911	IEQ = -2.384 + 0.463 × TC + 0.423 × VC + 0.529 × AC - 0.136 × IAQ
	Multivariate linear regression	0.908	IEQ = -3.106 + 0.438 × TC + 0.401 × VC + 0.525 × AC
	Arithmetic mean model	0.893	IEQ = -3.282 + 1.381 × (TC + VC + AC)/3
	Geometric mean model	–	IEQ = -2.656 + 1.306 × $\sqrt[3]{TC \times VC \times AC}$
[16]	Multivariate linear regression	0.89	IEQ = 0.736 × space + 0.063 × TC + 0.126 × VC
[31]	Multivariate linear regression	0.63	IEQ = 0.07 × IAQ + 0.20 × VC + 0.15 × AC + 0.12 × cleanliness and maintenance + 0.27 × space + 0.12 × Furnishing + 0.07 × colours and texture
[17]	Multivariate linear regression	–	IEQ = 0.12 × Temperature + 0.11 × Humidity + 0.12 × air movement + 0.49 × CO ₂ + 0.19 × PM _{2.5}
[26]	Multivariate logistic regression	–	IEQ = 1 - 1/(1 + exp(-15.0 + 6.09 × TC + 4.88 × IAQ + 3.7 × VC + 4.74 × AC))
[32]	Average IEQ scores in green building rating schemes and published papers	–	7 green building rating certificates IEQ = 0.27 × TC + 0.34 × IAQ + 0.22 × VC + 0.17 × AC
		–	9 publications IEQ = 0.28 × TC + 0.25 × IAQ + 0.21 × VC + 0.26 × AC

3.3.5. Updated IEQ acceptance model

One of the limitations of previously proposed IEQ models was the lack of flexibility to include additional IEQ parameters, e.g., privacy, cleaning and maintenance, vibration, movement and technology [11]. Also, the existing multivariate logistic regression models are not promising to predict occupant responses and acceptance, especially in poor IEQ conditions [51]. Hence, Wong et al. [11] proposed an open IEQ probabilistic acceptance model, which used frequency distribution functions of occupant's responses towards IEQ. They examined the acceptance of singular IEQ parameters and developed a model to predict the overall IEQ acceptance from the individual IEQ acceptance. Comparison between the performance of updated IEQ acceptance model with existing IEQ regression models [26,44,52] resulted in statistically similar prediction of overall IEQ acceptance rates. Tsang et al. [53] developed a Bayesian IEQ acceptance model based on the method proposed by Wong et al. [11]. They formulated the overall IEQ acceptance using separate binary responses of IEQ factors. It was determined that by updated IEQ acceptance model, the prediction accuracy can be acceptable even with a small quantity of field data.

Updated IEQ acceptance model can provide easier model updating based on the frequency distributions used. Also, this method is more robust in demonstrating occupants' psychological perception towards the indoor environment [11]. The updated IEQ acceptance model is flexible to various types and ranges of IEQ factors as well as new IEQ parameters.

3.4. IEQ models justification

Extracted papers used different methods to justify the reliability of their IEQ models. Some researchers compared the IEQ models with occupant satisfaction surveys and a few compared their results with national or international standards.

The simplest way to determine an occupant's satisfaction with an indoor environment is to ask them about their perception and satisfaction. Such studies are usually called post occupancy evaluation, and many researchers have conducted different surveys and presented their models to estimate IEQ. The key problem with such studies is the time dependency of surveys. A survey is conducted over a few days in a year (in the best case, a maximum of three different times of the year), and the whole model is developed from data collected in this limited survey period. For instance, due to a total dependency on occupant surveys, the linear model developed by Humphreys [12] failed to rank the indoor environments of the buildings in the correct order, as defined by the occupants' overall

assessments. UC Berkeley's Centre for the Built Environment survey database, which contains a large number of completed questionnaires (52,980 responses), was used in Heinzerling et al. [8]. Although occupants' satisfaction was regressed against overall workplace satisfaction, the absence of sufficient discussion and information on the accuracy and interpretability of proposed weighting was evident. Wong et al. [11] believed that the existing IEQ logistic regression model was not capable of predicting occupants' acceptance and therefore proposed a novel open IEQ acceptance model that assesses IEQ based on occupants' response frequency distribution of IEQ parameters. However, this IEQ acceptance models are completely dependent on user perception and satisfaction surveys over a limited period. Including objective measurement in the IEQ models may result in a better understanding of the IEQ condition in buildings.

Among the extracted papers, six compared their results with IEQ standards [8,9,17,20,34,35]. The standards used in these papers are presented in Table 5. EN-15251 is an international standard that includes all main IEQ factors (Thermal comfort, IAQ, visual comfort and acoustic comfort) [54–56]. This standard requires complicated IEQ data to calculate an overall IEQ indicator. EN-15251 and EN-16798-1 standards can be used for categorising buildings based on their energy consumption, ventilation, air quality, acoustics, and lighting [33].

Ncube et al. [9] introduced the assessment of IEQ performance with temporal variability as one of the key features of their evaluation method. They believed that the period of time a building falls into an assessment category is a more solid indicator of the comfort. An hourly criterion was employed to analyse the percentage of time that the criterion for specific category was met, and the result of this approach was compared with EN15251. Finally, two building ratings were developed accordingly.

Mujan et al. [23] implemented the EN 16798-1 category scheme and transformed it into ALDREN-TAIL Index. The main aim of this introduced index was to evaluate IEQ in the period of renovating existing buildings [23]. They used multivariate regression between overall satisfaction of indoor environment and main IEQ factors to determine each IEQ factor's relative weightings. Then, IEQ indexes were developed using EN 16798-1 standard to evaluate buildings performance during renovation [23].

3.5. The interaction between and relative importance of IEQ factors

As Heinzerling et al. [8] pointed out, few IEQ modelling studies tested the interaction effects between IEQ factors, nor included

Table 5

Standards used for IEQ model justification in the reviewed papers.

Establishment	Thermal comfort	IAQ	Visual Comfort	Acoustic Comfort
United States	ASHRAE 55	ASHRAE 62-1		
Europe	ISO 7730 EN 15251 EN 16798-1	EN 15251 EN 16798-1	EN 15251 EN 16798-1	ISO 3382 EN 15251 EN 16798-1
China	GB/T 50785	GB/T 18883	GB 50034	GB 50118
Singapore		Singapore 554	Singapore 531	

them in their IEQ models. Tang et al. [10] was the only extracted paper that has considered the interaction effects in their IEQ model. They found that both sound pressure level and illuminance had an impact on acoustic satisfaction; similarly, both the predicted mean vote and sound pressure level had an impact on perceived IAQ.

Apart from the interaction effects, Tang et al. [10] also reported that the impact of a specific IEQ factor on overall satisfaction fluctuated with its level relative to other IEQ factors. In other words, an environmental factor had a dominant influence on overall IEQ satisfaction when it had lower satisfaction compared with other factors. To ensure a good indoor environment, all aspects should consequently score highly enough. Being perfect in one or two IEQ aspects while neglecting other aspects can still lead to overall IEQ dissatisfaction.

Collecting surveys from 500 building occupants and physical measurements, Cao et al. [27] determined the occupant satisfaction with individual IEQ factors as a function of the related measured parameters. They used least-square multivariate linear regression to create a model for predicting overall IEQ satisfaction. Cao et al. [27] mentioned some limitations of the developed model. The model accuracy can only be guaranteed in roughly comfortable environments. Dissatisfaction with one extremely poor IEQ factor (e.g., being too cold or too noisy) will override satisfaction with other quality environmental parameters to reach a low overall IEQ satisfaction score, albeit the calculated results from their model predicted that occupants should be satisfied. This finding echoes Tang et al. [10] that (dis)satisfaction with the worst performing IEQ factor will dominate the overall IEQ (dis)satisfaction.

Table 6 demonstrates the ranking of IEQ factors' relative importance derived from studies using both objective measurement and subjective surveys, and studies which only relied on occupants' surveys. The consistency in the ranking when both measurement and surveys were included is evident in Ncube and Riffat [9] and Mujan et al. [23]. In both studies, IAQ was the most crucial factor and visual comfort had the least effect. On the contrary, there was a notable inconsistency in IEQ factors' relative importance from studies relying only on occupant surveys [20,22,31]. Tekce et al. [20] mentioned that the most and least influential factors in IEQ models were visual comfort and IAQ, respectively. However, Cheung et al. [22] obtained opposite results. They introduced IAQ as the most important IEQ factor and visual comfort as the least important in their developed IEQ model. According to Tang et al.

[10] and Cao et al. [27], this inconsistency in IEQ factor importance derived from subjective surveys may result from different "worst performing" IEQ factors in these studies.

4. Discussion

4.1. IEQ model evaluation criteria

Twenty-five reviewed studies are categorized based on the data collection sources (survey or measurement), analytical method (regression, average, AHP, or descriptive analysis) and number of survey participants in extracted papers, shown in Fig. 3. The number of publications in each category is also presented in parentheses. Among 25 reviewed papers, nine papers relied only on subjective surveys. While eight research studies distributed satisfactions surveys among building occupants [8,11,12,20,22,26,30,31], one study surveyed building experts instead [24]. Also, 13 studies combined IEQ measurement results with survey results [9,10,16,17,19,21,23,25,27-29,35,36]. While most researchers collected between 100 and 1,000 survey responses (13 papers) [10,11,16,17,19,20,22,23,25-27,29,35], three studies had responses less than 100 [9,21,24], and five previous papers gathered more than 1,000 surveys to establish an IEQ model [8,12,28,31,36]. Fas-sio et al. [45] claimed that smaller sample size in a POE study could cause unreliable results and hence proposed at least 300 satisfaction survey responses to achieve an optimal statistical reliability [45].

In evaluating the quality of the IEQ models, we believe that the number of IEQ factors included in the scoring system is crucial. As a minimum, it is not acceptable to develop an IEQ model neglecting one or more of the four main IEQ factors—thermal comfort, visual comfort, acoustic comfort, and IAQ [16,17]. As discussed in Section 3.5, the worst performing IEQ factor usually dominates occupants' overall IEQ satisfaction. IEQ evaluation concentrating on some aspects of the indoor environment while ignoring other important aspects may incur higher risks of overall IEQ dissatisfaction and greater threats to occupants' health, comfort, and productivity. Among the 25 studies assessed in the literature review, seven papers considered more than four IEQ factors [9,20,22,28,29,31,36], one investigated three factors [16] and another one included only two IEQ factors to determine the overall IEQ performance [17] (Fig. 4).

Table 6

Ranking of IEQ factors' importance in studies using subjective surveys only and studies employing both subjective and objective measurements (1 denotes most important, and 4 denotes least important).

Study	Year	Subjective surveys	Objective measurement	Rank			
				IAQ	Thermal comfort	Acoustic comfort	Visual comfort
[9]	2012	✓	✓	1	2	3	4
[23]	2021	✓	✓	1	2	3	4
[31]	2012	✓	–	3	4	1	2
[20]	2020	✓	–	4	3	2	1
[22]	2021	✓	–	1	3	2	4

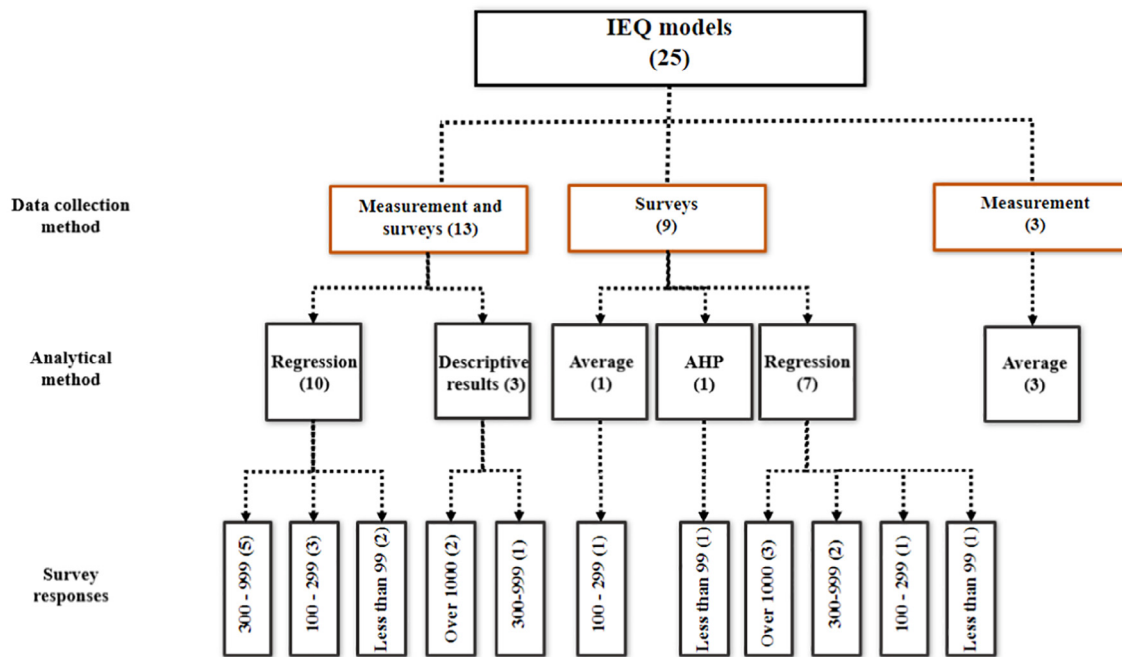


Fig. 3. Categorization of 25 studies on IEQ models by data collection method, analytical method, and number of survey responses.

Spatial variability of the measurement is another important criterion in developing quality IEQ models. In 16 studies that have carried out objective measurements, nine studies included at least two buildings for measurement sampling when developing IEQ models [9,19,23,25,27-29,35,36], and seven papers only carried out measurements in a limited space (room, floor or building) in their studies [10,16,17,21,32-34].

Similar to spatial variability of measurement, a third criterion is the temporal variability of measurement. Comparing the data collection period in 16 papers that have carried out physical measurements, nine studies measured the indoor environment for less than

a month [9,10,16,17,21,23,29,30], and seven measured the indoor environment for a whole year or in different seasons [16,21,25,27,28,34,36]. Obviously, developing an IEQ model based on a limited measurement period may lead to unreliable results.

A final criterion is the analytical method adopted to develop IEQ models. Regression analysis was the most favourable method for IEQ modelling and occupant satisfaction prediction among the 25 studies. In fact, eight papers reported a good determination coefficient (R^2 greater than 0.7) [8-10,16,21-23,26] and nine others either reported an R^2 less than 0.7 or did not report it [11,12,17,25,27,29-31,35]. In addition, four studies used a simple

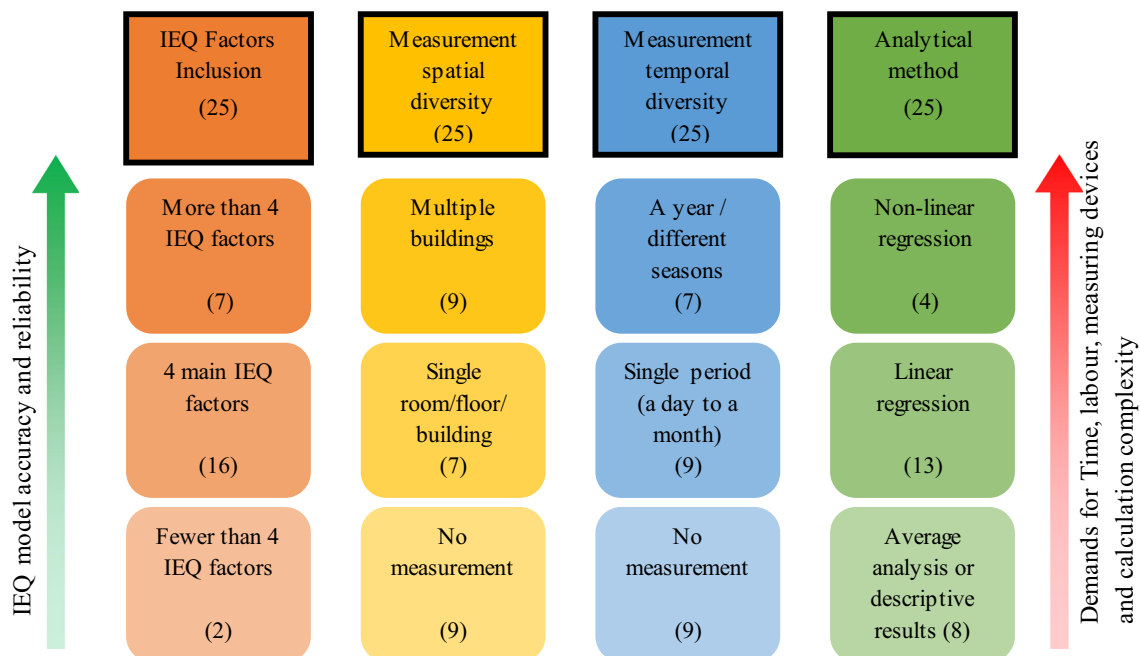


Fig. 4. Categorization of 25 studies in relation to key criteria that determine the accuracy and reliability of IEQ models.

Table 7

A simple ranking and scoring method for evaluating IEQ models.

	C (1 point)	B (2 points)	A (3 points)
Number of included IEQ factors P_{IIF}	Less than four	Four	More than four
Measurement temporal diversity P_{TD}	No measurement	Single period (a day to a month)	A year / different seasons
Measurement spatial Diversity P_{SD}	No measurement	Single room/ floor/ building	Multiple buildings
Analytical method P_{AM}	Average analysis or descriptive results	Linear regression	Advanced analysis (non-linear regression, model updating, etc)

arithmetic average procedure to develop IEQ models [20,32–34] or presented descriptive results only. Using simple analytical methods may cause inaccurate and unreliable IEQ models.

Type of IEQ models (subjective, objective and subjective-objective models) may potentially affect the accuracy and reliability of IEQ models as well, as has been discussed in Section 3.5. However, more research studies are needed in each type before a robust conclusion can be made regarding the model accuracy and reliability.

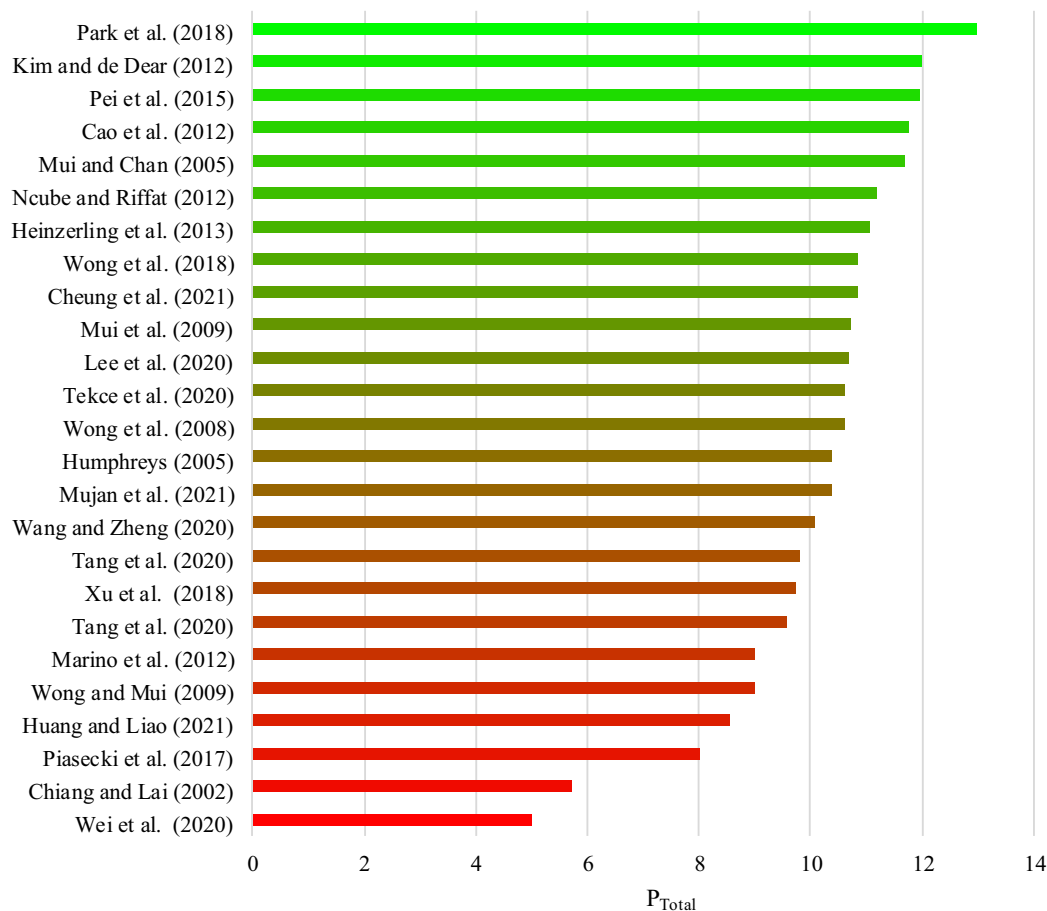
4.2. A simple ranking and scoring method

Based on above discussion, the IEQ models developed in 25 reviewed studies are assessed against five evaluation criteria that affect the quality of the IEQ models. The first aspect is the number of included IEQ factors (P_{IIF}). Temporal diversity (P_{TD}) and spatial diversity (P_{SD}) are aspects pertaining to the physical data collection. The next aspect is the analytical method (P_{AM}) used for developing IEQ models. The last evaluation aspect, which was not mentioned before, is the number of questionnaire responses (P_{SR}) if subjective measurement was employed. The total points representing the overall quality of the developed IEQ model (P_{Total}) is calculated using Eq. (11).

$$P_{Total} = P_{IIF} + P_{TD} + P_{SD} + P_{AM} + P_{SR} \quad (11)$$

A simple ranking and scoring method is developed for evaluating the overall quality of established IEQ models in 25 studies. Table 7 introduces how four aspects (P_{IIF} , P_{TD} , P_{SD} , P_{AM}) are evaluated using three classes of ranking and scoring system (C-1 point, B-2 points, A-3 points).

Using logarithmic scaling, Eq. (12) normalizes the number of survey responses compared to the maximum number of surveys (MNS) in a publication among 25 extracted papers (52,980 responses) and calculates the related score for each study. P_{SR} is considered equal to zero for studies without survey responses and equal to one for studies with more responses than MNS.

**Fig. 5.** Ranking of IEQ models developed in extracted studies.

$$P_{SR} = \begin{cases} 0 & MNS = 0 \\ 3 \times \frac{\log(\text{number of survey respondents})}{\log(MNS)} & 0 < MNS \leq 52,980 \\ 1 & 52,980 < MNS \end{cases} \quad (12)$$

Fig. 5 illustrates the total points (P_{Total}) of IEQ models in 25 reviewed studies. Two studies (Park et al., 2018; Kim and de Dear, 2012) achieved a total point no less than 12 (80% of the total score), and 16 studies had a total point above 10 (60% of total score). As illustrated in Fig. 5, the study by Park et al. [36] received the highest rank. This study included physical measurement in 64 office buildings during heating, cooling and swing seasons and collected 1,601 survey responses. Collection of wide-ranging data from both objective and subjective measurements in multiple buildings and multiple seasons provides a comprehensive evaluation of the buildings' IEQ performance and therefore the developed IEQ models should be more accurate and reliable. This study can be used as an exemplary case for future studies on IEQ model development.

5. Conclusion

IEQ models combine different IEQ factors into an individual overall indicator. Comparing previously developed IEQ models was a challenge due to diverse measurement protocols, IEQ weighting schemes and analytical methods, which illustrates the importance of IEQ assessment benchmarking. This paper comprehensively reviews 25 published studies aiming to develop IEQ evaluation models in commercial buildings. Main results are summarized below:

After Heinzerling's review [8], the open IEQ acceptance model [11] emerged as a new analytical method to develop IEQ models that enable the inclusion of additional IEQ parameters, model updating with the frequency distribution data, and improved prediction of occupants' psychological perceptions.

The interaction effects between IEQ factors and relative contribution of them to the overall IEQ satisfaction were also investigated in recent studies. The worst performing IEQ factors will usually dominate occupants' overall IEQ satisfaction. IEQ factors' relative contribution developed from both objective measurements and subjective surveys display greater consistency than that developed from subjective surveys only.

Based on the review of extracted 25 research papers, five key criteria for developing quality IEQ models—number of IEQ factors included, spatial and temporal variability in physical measurement, analytical methods for model development, and the number of responses if subjective surveys are carried out—are proposed and discussed.

A simple ranking and scoring tool is developed for evaluating the quality of the IEQ models proposed in the reviewed studies. Results show that two studies achieved a total score no less than 80% of the total score, and 16 studies achieved a minimum of 60% of the total score. While inclusion of extra IEQ factors, extended measurement and more complicated analysis can demand extra time, labour, measurement devices, etc., neglecting these aspects can lead to loss of accuracy and reliability. This simple tool can help researchers quickly evaluate IEQ models developed from different methods.

CRedit authorship contribution statement

Soheil Roumi: Writing – original draft, Methodology, Investigation, Formal analysis. **Fan Zhang:** Conceptualization, Formal analysis, Writing – review & editing. **Rodney A. Stewart:** Writing – review & editing, Methodology, Supervision. **Mattheos Santamouris:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that there is no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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